

Motor Sequence Learning Experiment

PSY310 Lab in Psychology

Lab Report

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GitHub Link :

Introduction

Motor Sequence learning (MSL) is an experimental archetype used to examine how procedural learning (underlying skill from MSL) is used to acquire and process different sequences of motor actions. Procedural Learning is a form of memory which differentiates distinct memory from knowledge. It allows us to perform our skills smoothly with very little usage of conscious thoughts. Initially there are three parts of the brain that are active during the learning - the Prefrontal Cortex, Basal Ganglia and the Cerebellum. They work together in providing conscious attention and planning, formation of habits and taking care of timing, coordination and accuracy (respectfully). Learning signifies as a permanent and more dominant change in the behavior whilst Contingency refers to the as a statistical and relational concept, between two events. The underlying principle of contingency is what makes associative learning possible, for example - contingency helps to predict outcomes and adjust their behaviors and an inconsistent relationship results in poor learning.

Method

Participants

The experiment was performed by five participants; all the participants were 20 years old and female undergraduate students studying at Ahmedabad University. All of them were aware of the experiment's objective, i.e., to study how individuals retain and incorporate sequence of actions.

Materials & Procedure

The experiment was created on a laptop, using the PsychoPy v2025.1.1. 200 trials were conducted per participant.

The experiment had five primary components - a fixation cross; to attention and also so that the participants understood it as a visual cue. Then we added a rectangle polygon (Line 1) , once it was added - we added the three more similar polygons (Line 2, Line 3 and Line 4) only changing their positions for each line. We added a triangle probe, which would be situated on top of the rectangle lines (Figure 1). And lastly we added the key response.

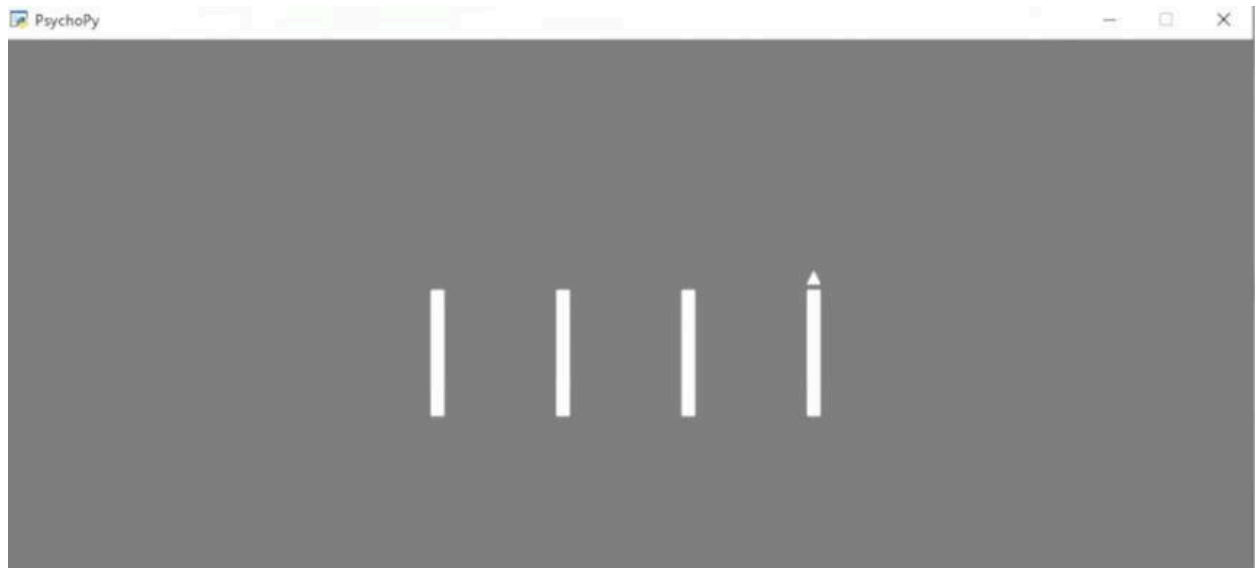


Figure 1

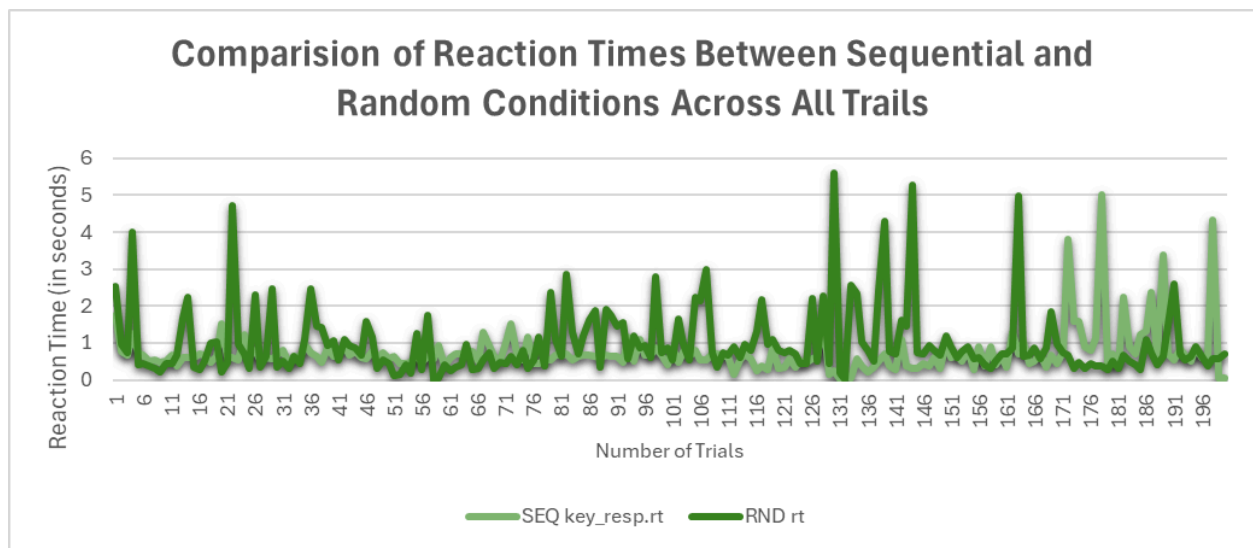
After that, we inserted two loops - the first loop type was sequential and the second loop type was random. Conditions for both the files were added through a .csv file; where each position corresponds to a key press (Figure 2).

	A	B
1	pos	corr_resp
2	-50	x
3	150	v
4	50	c
5	-150	z
6		
7		

Figure 2

After this, we run the experiment and the data will be automatically saved in an excel file.

Results



	sq.rt	rnd.rt
average	0.73536	0.99267
difference	0.25731	

The mean difference in reaction times (RT) between the two conditions was 0.26 seconds, with sequential trials being faster. This signifies that the participants were able to respond faster - by understanding the sequence.

Discussions

The difference between the RTs for both conditions reflects on Motor Sequence Learning, i.e., where repeated exposure to a sequence enhances efficiency. The participants were able to pick up on the patterns, which reduced the cognitive load and produced answers quickly. In the random condition, each trial required a new stimulus planning - which led to the slower RTs. However there is one exception; Participant Five, although the difference between the reaction times is 0.09 seconds with Random Condition being faster the answers for the Sequential Condition were all accurate. Counterbalancing the conditions would lead to a bias in the results. But by balancing the conditions here, we were able to identify the differences in RTs; because of condition type (sequential VS random) and not by factors such as, general practice or order effects or in the way that they were experienced. Overall, this would strengthen the internal validity of the study.

References

1. Kantowitz, B. H., Roediger, H. L., & Elmes, D. G. (2009). *Experimental psychology*. Boston, MA, USA: Wadsworth Cengage Learning.